

‘**Erythrophylla**’ see *M.I.* ‘Erythroneura’. ‘**Kerchoveana**’ (Rabbit’s tracks) illus. p.475. ‘**Massangeana**’ is an evergreen, short-stemmed perennial, branching at the base. H and S 1ft (30cm). Each large, oblong, velvety, dark green leaf has a wide, irregular, pale midrib, white, lateral veins, is often purplish-green below; they stand upright at night, but lie flat during the day. Small, white to mauve flowers are produced in slender, upright spikes year-round. Z12 (min. 59°F/15°C)

Marginatocereus marginatus. See *Pachycereus marginatus*.

MARGYRICARPUS

ROSACEAE

Genus of dwarf, evergreen shrubs, grown for their fruits. Good for rock gardens. Frost hardy. Needs a sheltered, sunny position and well-drained soil. Propagate by softwood cuttings in late spring or early summer or by seed in fall. **M. pinnatus**, syn. *M. setosus* (Pearl berry). Evergreen, prostrate shrub. H 9–12in (23–30cm), S 3ft (1m). Dark green leaves are divided into linear, silky leaflets. In early summer, produces inconspicuous, green flowers, then small, round, glossy, purple-tinted, white fruits. Z8–10 to Z9–11 **M. setosus.** See *M. pinnatus*.

MARTYNIA

MARTYNIACEAE

Genus of annuals, grown for their tubular, pale pink flowers and horned fruits. Half hardy. Requires a sunny, sheltered site and fertile, well-drained soil. Propagate by seed sown under glass in early spring. **M. annua** (Unicorn plant) illus. p.300.

MASDEVALLIA

ORCHIDACEAE

See also ORCHIDS. **M. coccinea** (illus. p.466). Evergreen, terrestrial orchid for a cool greenhouse. H 6in (15cm). Has oblong to lance-shaped leaves, to 8in (20cm) long. Bears white and yellow, red-orange, crimson, or purple-pink flowers, 3in (8cm) long, in summer. Needs shade in summer. Z11–12 (min. 52°F/11°C; max. 75°F/24°C) **M. infracta.** Evergreen, epiphytic orchid for a cool greenhouse. H 6in (15cm). Has oblong to lance-shaped leaves, to 5½in (14cm) long. Bears rounded, yellow-flushed, red to purplish-pink flowers, 2in (5cm) long, with taillike, greenish sepals, in summer. Requires a position in shade from hot summer sun. Z11–12 (min. 52°F/11°C; max. 75°F/24°C) **M. towarensis** (illus. p.466). Evergreen, epiphytic orchid for a cool greenhouse. H 6in (15cm). Has oblong to lance-shaped leaves, to 6in (15cm) long. In winter, milky-white flowers, 1½in (4cm) long, with short-tailed sepals, singly or up to 3 to a stem. Grow in shade in summer. Z11–12 (min. 52°F/11°C; max. 75°F/24°C) **M. wagneriana** (illus. p.467). Evergreen, epiphytic orchid for a cool greenhouse. H 3in (8cm). Narrowly ovate leaves are 4in (10cm) long. Produces pale yellow flowers, 1½in (4cm) long, with taillike sepals, singly

or in pairs, in summer. Needs a position in shade in summer. Z10–11

MATHIASELLA

APIACEAE

Genus of one species of perennial, grown for its umbels of bell-shaped, green flowers. Fully hardy. Needs sun and soil that is well-drained, especially in winter. **M. bupleuroides ‘Green Dream’** illus. p.242.

MATTEUCCIA

ONOCLEACEAE

Genus of deciduous, rhizomatous ferns. Fully hardy. Prefers partial shade and moist soil. Remove faded fronds regularly and divide plants when crowded. Propagate by division in early spring. **M. orientalis.** Deciduous, rhizomatous fern. H and S to 3ft (1m). Has a “shuttlecock” of sterile, arching, broadly ovate, divided fronds, to 32in (80cm) long, light green when young, becoming darker. Fertile, erect, blackish-green fronds emerge from the center of the plant in summer. Z4–8 **M. struthiopteris** (Ostrich-feather fern, Ostrich fern) illus. p.443.

MATTHIOLA

Gillyflower, Stock

BRASSICACEAE/CRUCIFERAE

Genus of bushy, erect annuals, biennials, and perennials, occasionally subshrubs. Flowers of most annual or biennial stocks are highly scented and excellent for cutting. Fully hardy to frost hardy. Grow in sun and in fertile, well-drained, ideally lime-rich soil. Tall cultivars may need support. If grown as biennials outdoors, provide cloche protection in winter. For flowers outdoors the same summer, sow seed of annuals under glass in early spring, or outdoors in mid-spring. Sow seed of perennials under glass in spring or summer. Propagate subshrubs by semiripe cuttings in summer. Prone to aphids, flea beetle, club root, downy mildew, and botrytis. **M. bicornis.** See *M. longipetala* subsp. *bicornis*. **M. incana** (Gillyflower, Stock). Fast-growing, upright, bushy subshrub, or short-lived perennial, grown as an annual. H 1–2ft (30–60cm), S 1ft (30cm). Has lance-shaped, grayish-green leaves and, in summer, scented, fragrant, mauve, purple, violet, pink, or white flowers borne in spikes. (borderline) Z8–10. **Cinderella Series** cultivars, H 8–10in (20–25cm), S to 10in (25cm), have double, dark blue-purple, lavender-blue, red, rose-pink, silvery-blue, or white flowers. Flowers of **Excelsior Mammoth Column Series** cultivars, H to 30in (75cm), S to 1ft (30cm), are mostly double, pink, red, pale blue, or white. **Giant Excelsior Mix** illus. p.303 **Hot Cakes Mixed** cultivars, S 6–8in (15–20cm), are upright, with spikes of mainly double, white, cream, pink, rose-pink, or purple flowers. Cultivars in **Sentinel Series**, H to 30in (75cm), S to 1ft (30cm), have double flowers in hues including white, pink, carmine-red, and light to dark blue. **Sugar and Spice Mixed** cultivars are compact and bushy, with spikes of

mainly double, white, pink, violet, or yellow flowers. **Ten Week Mixed** cultivars, H 1ft (30cm), S to 10in (25cm), bear mainly double flowers in shades of crimson, pink, lavender-pink, purple, and white. **Vintage Mixed** is sturdy and branching. H 16in (40cm), S 1ft (30cm), bearing spikes of single or double flowers in burgundy red, copper orange, lavender, red, white, yellow, and rich crimson pink. **M. longipetala subsp. bicornis**, syn. *M. bicornis* (Night-scented stock). Erect to spreading, single-stemmed to branching annual. H 12–14in (30–35cm), S 9in (23cm). Has narrow, sometimes divided, gray-green leaves. In summer, produces open spikes of pink, mauve, or purple flowers that are strongly fragrant at night. Z9–10

MATUCANA

CACTACEAE

Genus of low-growing, spherical to shortly cylindrical, solitary to clustering, perennial cacti, with thick, ribbed stems, often with some spines, usually branching from the base. Solitary, narrowly funnel-shaped yellow, orange or red flowers are produced around the stem tips in summer. Frost tender. Needs full sun and very well-drained, slightly alkaline soil. Propagate by seed in spring or summer. **M. aurantiaca**, syn. *Oreocereus aurantiacus*. Very variable, spherical to shortly cylindrical, perennial cactus. H 5in (12cm), S 16in (40cm). Has a 16-ribbed stem and elongated areoles each bear up to 25 spines. Bears orange-red or orange-yellow flowers in summer. Z10–11 (min. 41°F/5°C) **M. haynei**, syn. *Oreocereus haynei*. Slow-growing, spherical to columnar, perennial cactus. H 2ft (60cm), S 4in (10cm). Produces a cylindrical, much-ribbed, grass-green stem, densely covered with short, white or yellowish-brown spines. Bears orange-brown, red, or purple-crimson flowers in summer on plants more than 6in (15cm) tall. Z10–11 (min. 41°F/5°C)

MAURANDYA

PLANTAGINACEAE/SCROPHULARIACEAE

Genus of twining, woody-based, perennial climbers, grown against a wall or to cover a trellis. Half hardy. Needs full sun and moderately fertile, moist but well-drained soil. Propagate by seed in spring, or softwood cuttings in late spring. **M. barclayana**, syn. *Asarina barclayana*. Evergreen, soft-stemmed climber, herbaceous in cold climates. H to 6ft (2m). Has angular, heart-shaped, hairless leaves. Trumpet-shaped, white, pink, or purple flowers, each with a green or whitish throat, 2½–3in (6–7cm) long, are produced from summer to fall. Z10–11 **M. erubescens.** See *Lophospermum erubescens*.

MAXILLARIA

SYN. BRASILIORCHIS, MAXILLARIELLA
ORCHIDACEAE

See also ORCHIDS. **M. picta**, syn. *Brasiliorchis picta*. Evergreen, epiphytic orchid for a cool greenhouse. H 9in (23cm). Has one narrowly ovate leaf, 6–9in (15–23cm) long. Fragrant, deep yellow to white flowers,

marked purple to dark reddish-brown outside, are borne singly beneath foliage in winter. Requires partial shade in summer. Z11–12 (min. 50°F/10°C; max. 86°F/30°C) **M. porphyrostele**, syn. *Brasiliorchis porphyrostele* (illus. p.467). Evergreen, epiphytic orchid for a cool greenhouse. H 3in (8cm). Has narrowly ovate leaves, 3in (8cm) long. Slightly fragrant, white- and red-lipped, yellow flowers, with purple-striped throats, are borne singly in summer and fall. Grow in good light in summer. Z10–11 (min. 50°F/10°C; max. 86°F/30°C) **M. tenuifolia.** Evergreen, epiphytic orchid for a cool greenhouse. H 6in (15cm). Has narrowly ovate leaves, 6in (15cm) long. Fragrant, yellow flowers, heavily overlaid with red and with white lips, are borne singly throughout summer. Needs good light in summer. Z11–12 (min. 50°F/10°C; max. 86°F/30°C)

Maxillariella. See *Maxillaria*.

MAYTENUS

CELASTRACEAE

Genus of evergreen trees and shrubs, grown for their neat foliage. Fully hardy to frost tender, but needs shelter from strong, cold winds when young. Requires sun or partial shade and fertile, well-drained soil. Propagate by semiripe cuttings in summer or by suckers in spring. **M. boaria**, syn. *M. chilensis* (Maiten). Evergreen, bushy-headed, elegant tree or shrub. H 70ft (20m), S 25ft (8m). Produces narrowly ovate to lance-shaped, glossy, dark green leaves on slender shoots, and tiny, star-shaped, green flowers in late spring. Z7–10 **M. chilensis.** See *M. boaria*.

MAZUS

PHRYMACEAE/SCROPHULARIACEAE

Genus of annuals and creeping, spring-flowering perennials. Useful for rock gardens and in paving. Borderline hardy to fully hardy. Needs a sheltered, sunny site and moist soil. Propagate by division in spring or by seed in spring or fall. **M. reptans** illus. p.351.

MECONOPSIS

PAPAVERACEAE

Genus of annuals, biennials, and deciduous or evergreen perennials, some long-lived, some short-lived and others monocarpic (dying after flowering), grown for their flowers. Fully hardy. Needs shade and, in warm areas, a cool position. Most prefer humus-rich, moist, neutral to acidic soil. *M. baileyi* and *M. ‘Crewdson Hybrid’* tolerate more alkaline soil. Propagate sterile perennial hybrids and perennial species by division in late summer or early spring. Propagate fertile species and hybrids (which set viable seed) by seed in spring. Division of perennial species and cultivars is advisable every 3 years. **M. baileyi**, syn. *M. betonicifolia* of gardens (Himalayan blue poppy, Tibetan blue poppy). Clump-forming, deciduous perennial. H 3–4ft (1–1.2m), S 1½ft (45cm). Oblong to ovate, hairy, mid-green leaves, with heart-shaped bases, are produced in